



Maintenance Schedule

Diesel Engine-Generator Sets
with 12V/16V/18V2000Gx5 engines
Application groups 3B, 3D

MS50117/03E

System designation	Engine model	Hours	Application group
DP00770D5S	12V2000G65-TD	10,000h	3B, continuous operation, variable load, ICXN
DP00820D5S	12V2000G65-TB	10,000h	3B, continuous operation, variable load, ICXN
DP00900D5S	16V2000G25-TD	10,000h	3B, continuous operation, variable load, ICXN
DP00910D5S	16V2000G25-TB	10,000h	3B, continuous operation, variable load, ICXN
DP01000D5S	16V2000G65-TD	10,000h	3B, continuous operation, variable load, ICXN
DP01030D5S	16V2000G65-TB	10,000h	3B, continuous operation, variable load, ICXN
DP01120D5S	18V2000G65-TD	10,000h	3B, continuous operation, variable load, ICXN
DP01130D5S	18V2000G65-TD	10,000h	3B, continuous operation, variable load, ICXN
DP01180D5S	18V2000G65-TB	10,000h	3B, continuous operation, variable load, ICXN
DS00850D5S	12V2000G65-TD	6,000h	3D, Emergency Standby power, IFN
DS00890D5S	12V2000G65-TB	6,000h	3D, Emergency Standby power, IFN
DS01000D5S	16V2000G25-TD	6,000h	3D, Emergency Standby power, IFN
DS01050D5S	16V2000G25-TB	6,000h	3D, Emergency Standby power, IFN
DS01100D5S	16V2000G65-TD	6,000h	3D, Emergency Standby power, IFN
DS01140D5S	16V2000G65-TB	6,000h	3D, Emergency Standby power, IFN
DS01250D5S	18V2000G65-TD	6,000h	3D, Emergency Standby power, IFN
DS01290D5S	18V2000G65-TB	6,000h	3D, Emergency Standby power, IFN

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Maintenance schedule matrix 10,000 h

0-10,000 Operating hours

Item	Limit	Operating hours [h]																					
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	
Engine-generator set monitoring																							
Engine-generator set monitoring	-	X																					
Engine-generator set monitoring	1 m																						
Engine																							
Engine oil filter	2 a																						
Fuel filter	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Valve gear	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Fuel prefilter	2 a			X		X		X		X		X		X		X		X		X		X	
Fuel injectors	-							X						X						X			
Pressure pipe neck (bayonet)	-							X						X						X			
Belt drive	2 a							X						X						X			
Air filters	3 a							X						X						X			
Combustion chambers	4 a							X						X						X			
Fuel pressure maintaining valve	-													X									
Fuel injection pumps	-													X									
Turbochargers	9 a																				X		
Component maintenance	9 a																				X		
Cylinder heads	9 a																				X		
Extended component maintenance	18 a																					X	
Generator																							
Generator condition	1 m																						
Wiring	1 a																						
Circuit breaker	1 a																						
Antifriction bearing	1 a						X					X					X				X		
Generator	18 a																					X	
Automatic transfer switch ATS																							
ATS	1 m																						
ATS	1 a																						
Preheater 2,5 kW																							
Preheater	5 a																						
Preheater 6 kW																							
Control box	1 a																						
Heating element housing	1 a																						
Heating element and heating element housing	1 a																						
Thermostat	5 a																						
Solenoid switch	10 a																						
Engine-generator set control panel																							
Backup battery	10 a																						
Cooler / Radiator																							
Cooler / Radiator	6 m		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Cooler / Radiator	18 a					X				X				X			X					X	
Engine-generator set																							
w = weeks m = months a = years																							

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Item	Limit	Operating hours [h]																				
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000
Rubber sleeves	6 a										X											X
Hose lines	6 a			X		X		X		X		X		X		X		X		X		X
w = weeks m = months a = years																						